

Short literature scan – sEMG gesture recognition with Ninapro DB1

1. Context

Our project uses **Ninapro DB1**, specifically exercise E1 (12 finger movements), with **10 surface-EMG channels**. DB1 contains recordings from 27 intact subjects, with 10 repetitions of each movement. The first eight electrodes are distributed around the forearm and two additional electrodes target the flexor and extensor digitorum superficialis. Importantly, the OttoBock electrodes used in DB1 do not provide raw bipolar EMG: they output an **amplified, filtered and RMS-rectified version of the EMG signal**. This matters when deciding which “standard EMG features” are actually meaningful for our data. [1]

The main problem we already expect is the large difference between **within-subject classification** and **new-subject classification**. EMG patterns depend not only on the gesture, but also on anatomy, muscle geometry, electrode position, skin/fat properties, contraction intensity and other subject-specific factors. Therefore, a classifier can learn one person quite well while generalising badly to an unseen person. This is a well-established limitation of myoelectric pattern recognition.

2. Classical features are still a strong starting point

One of the classic papers in EMG pattern recognition is Hudgins et al. (1993), which introduced a compact time-domain representation for multifunction myoelectric control. The well-known “Hudgins” family includes features such as mean absolute value, waveform length, zero crossings and slope-sign changes. The main advantage is that these features are computationally cheap and capture amplitude and temporal complexity without needing a complex spectral model. [2]

However, **we cannot blindly copy the complete Hudgins set**. Zero crossings and slope-sign changes are useful on bipolar raw EMG because polarity changes carry information. DB1 is already RMS-rectified, so those features lose their original physiological meaning. For our dataset, amplitude/energy-like features such as **MAV, RMS, variance and waveform length** are much more defensible.

This is supported directly by the Ninapro benchmark study by Atzori et al. The authors compared multiple feature representations and classifiers and found that relatively simple features such as **MAV and waveform length performed similarly to more computationally demanding wavelet representations**. They also found that a **nonlinear SVM** was consistently strong across different feature representations. [3]

This suggests that our supplied baseline is not artificially weak: simple time-domain features are already established choices for Ninapro. Therefore, our project should probably not aim

at adding dozens of arbitrary features. A more interesting question is whether we can construct features that are **less dependent on the absolute amplitude of a particular subject**.

Phinyomark et al. also studied a large set of time- and frequency-domain EMG features and showed substantial redundancy between many commonly used features. Their conclusion is relevant for us: **more features do not automatically mean more information, and feature selection or a compact, motivated feature set can be preferable to simply concatenating everything available**. [4]

3. Why cross-subject recognition fails

A major issue in practical EMG recognition is **electrode shift**. Even relatively small changes in electrode position alter which motor units and muscles contribute to each channel. Young, Hargrove and Kuiken experimentally showed that pattern-recognition performance degrades when electrodes shift relative to the underlying muscles. They also showed that electrode configuration and the choice of features affect robustness to this shift. [5]

This is closely related to our new-subject problem. In Ninapro, the nominal electrode layout is consistent, but the same channel does not necessarily measure an identical physiological source in every subject. Different forearm dimensions and anatomy effectively produce a spatial/domain shift.

This means that features such as

$$\text{MAV}(\text{channel } 4) = 0.7$$

may be highly informative within one subject but are not necessarily comparable to the same numerical value for another subject.

For this reason, one promising direction is to represent the **relative activation pattern across channels, rather than only absolute channel amplitudes**. For example, channel activation could be normalised relative to total activation across the forearm. This is not directly prescribed by the papers above, but it follows from the documented cross-user and electrode-position problem and is an experimentally testable hypothesis for our project.

4. Normalisation seems particularly relevant

There is direct evidence that normalisation can improve subject-independent EMG classification.

Wahid et al. (2018) investigated subject-independent gesture recognition and compared original time-domain EMG features with normalised versions. They normalised feature values using muscle-contraction energy (AUC of the RMS signal), and reported improved subject-independent performance compared with unnormalised features. Their study used only three gestures and a different acquisition setup, so the exact method cannot simply be

copied to Ninapro, but the result supports the broader idea that **removing subject-specific amplitude scaling can improve generalisation**. [6]

Lin et al. (2020) reached a similar conclusion using a ConvNet. They proposed a normalisation procedure intended specifically for **inter-subject sEMG recognition**, using information from a short calibration cycle of the new user to align amplitude ranges. Their results improved cross-subject recognition substantially on their datasets. [7]

For us this creates two different experimental questions:

- Can we obtain better cross-subject performance using a **completely subject-independent normalisation**, requiring no information from the test subject?
- If we use statistics from the new subject, are we instead defining a **calibrated deployment scenario**?

This distinction is important. Using statistics calculated from the complete held-out subject before classification is no longer the same claim as “zero-calibration recognition of an unseen subject”. Any such procedure must be explicitly defined and evaluated without leakage.

5. Domain adaptation: effective, but probably a later step

More recent work treats each subject as a different domain and explicitly performs **domain adaptation**.

Wang et al. (2023) proposed an iterative self-training domain-adaptation method for cross-user sEMG recognition. The method aligns source and target feature distributions and iteratively assigns pseudo-labels to unlabeled target-user data. Importantly for us, they tested the approach on public datasets including **Ninapro DB1 and DB5**, and reported improvements over conventional cross-user approaches. [8]

This confirms that the cross-subject gap is not simply a matter of choosing a better classifier: the **distribution itself changes between users**.

However, domain adaptation is probably not the best first modification for our project. It introduces substantially more complexity and often assumes access to unlabeled data from the target user. A simpler progression would be:

baseline → normalisation / relative features → model comparison → only then consider adaptation if necessary.

This gives a much clearer experimental story and lets us identify which improvement is responsible for any gain.

6. What this literature suggests we should test

Based on this short scan, the most defensible initial direction seems to be:

1. **Keep the supplied MAV/RMS/WL/VAR-based representation as our baseline.**
2. Avoid raw-EMG features such as zero crossings and slope-sign changes because DB1 is already RMS-rectified.
3. Test whether **normalisation** reduces subject-specific amplitude differences.
4. Test **channel-relative features**, i.e. representing the spatial activation pattern relative to total or mean forearm activation.
5. Compare a small number of classifiers. A nonlinear **SVM** is particularly justified by the original Ninapro benchmark; Random Forest remains a useful baseline.
6. Analyse performance separately for within-subject and new-subject evaluation. Improvement only in within-subject performance would not solve the main generalisation problem.
7. Treat domain adaptation/transfer learning as a possible later experiment rather than the starting point.

Deep neural networks are widely used in current EMG research, but the literature does not imply that “deep = automatically better”. For our project, a simpler model with carefully designed, physiologically meaningful and subject-robust features may produce a more interpretable experiment. The important question is not only whether the final macro-F1 increases, but **why** it changes and whether the improvement survives evaluation on unseen subjects.

References

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